

SOME LIMIT THEOREMS FOR STATIONARY PROCESSES

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Introduction

In the present paper, we establish a series of limit theorems for sums of the form

$$(1) \quad S_n = \frac{x_1 + x_2 + \cdots + x_n}{B_n} - A_n,$$

where $\{x_j\}$, $-\infty < j < \infty$, is a process which is stationary in the strict sense, while A_n and B_n are normalizing constants. Evidently, interesting results in this direction can be obtained only if we impose certain restrictions on our process, which, roughly speaking, consist in the requirement that the dependence between the beginning and end of the process be weak. In particular, we consider processes satisfying one of the following requirements: for all $B \in \mathfrak{M}_{k+n}^\infty$ with probability 1

$$(I) \quad |\mathbf{P}(B|\mathfrak{M}_{-\infty}^k) - \mathbf{P}(B)| \leq \varphi(n) \downarrow 0, \quad n \rightarrow \infty,$$

and

$$(II) \quad \sup |\mathbf{P}(AB) - \mathbf{P}(A)\mathbf{P}(B)| \leq \alpha(n) \downarrow 0, \quad n \rightarrow \infty.$$

Here the supremum is taken over all $A \in \mathfrak{M}_{-\infty}^k$ and $B \in \mathfrak{M}_{k+n}^\infty$, $\mathfrak{M}_a^b$ denotes the σ -algebra generated by events of the form $\{(x_{i_1}, \cdots, x_{i_k}) \in E\}$, where $a \leq i_1 < i_2 < \cdots < i_k \leq b$, and E is a k -dimensional Borel set. Condition (I) (considered by the author in [13]) is essentially the extension to arbitrary processes of the ergodicity coefficient introduced by R. L. Dobrushin for Markov processes (see [8]).

Condition (II) was first used by Rosenblatt in [26] (see also [5], [13] and [25]). Following Rosenblatt, we shall call (II) the *strong mixing condition*.

Examples of processes satisfying condition (I), besides independent random variables, are m -dependent random variables [7], which are Markov processes satisfying Doeblin's condition. Less familiar examples are certain chains of infinite order¹ [see e.g., [23], [10], [12]], which have recently been employed in learning theory (see [3], [19]). Still another interesting example can be found in a paper by the author [14]. Examples of processes satisfying condition (I) (but not satisfying condition (II)) can easily be obtained by using the results of the notes [16] and [18].

¹ The corresponding French term is *chaînes à liaisons complètes*.

In Section 1, we show that in the case of processes satisfying conditions (I) and (II), the sums of the form (1) can only converge to stable laws. In this section, we give a series of theorems concerning the convergence of distribution functions of the sums (1) to a normal law. In particular, we note that Theorem 1.5 of Section 1 contains the results of Nagaev on the asymptotic normality of sums of random variables which are connected in a homogeneous Markov chain and the results of Diananda on the asymptotic normality of sums of stationary m -dependent random variables [7]. The results of Sections 1.4–1.5 are very closely related to results obtained by Yu. A. Rozanov (see the first part of [5] and [25]).

It should be noted that investigation of the conditions under which the central limit theorem can be applied to processes satisfying conditions (I) and (II) essentially means a return to the investigations of S. N. Bernstein (see e.g., [1]) on the conditions for asymptotic normality of sums of weakly dependent random variables.

In Section 2 we investigate processes $\{f_j\}$ of the form

$$\begin{aligned} & \dots\dots\dots, \\ f_{-1} &= f(\dots, x_{-2}, x_{-1}, x_0, \dots), \\ f_0 &= f(\dots, x_{-1}, x_0, x_1, \dots), \\ f_1 &= f(\dots, x_0, x_1, x_2, \dots), \\ & \dots\dots\dots, \end{aligned}$$

where $\{x_j\}$ is a process satisfying conditions (I) or (II) (cf. [2], [14], [15]).

Throughout the entire paper all random events and random variables are considered over a basic probability space $\{\Omega, \mathfrak{F}, \mathbf{P}\}$, whose elements are denoted by ω . Thus for example, if x is a random variable with a mathematical expectation, then

$$\mathbf{E}x = \int_{\Omega} x \mathbf{P}(d\omega) = \int_{\Omega} x(\omega) \mathbf{P}(d\omega).$$

As already noted, we confine ourselves to the study of processes which are stationary in the strict sense, and hence the words “in the strict sense” will usually be omitted from now on. Every stationary process $\{x_j\}$ induces a probability-conserving transformation of events, which will be denoted by T , on the σ -algebra of events $\mathfrak{M}_{-\infty}^{\infty}$ generated by the process. Moreover, we shall denote by T_1 the transformation of random variables, measurable with respect to $\mathfrak{M}_{-\infty}^{\infty}$, which is generated by the stationary process $\{x_j\}$ (see e.g., [11], Chap. 10).

By the letter M , we shall denote any quantity (not always the same) which is bounded in absolute value, so that inequalities and equalities of the following kind are possible:

$$\sum_{l=2}^{n-2} \frac{(n-l)^2}{\log l} \leq M n^2 \sum_{l=2}^{n-1} \frac{1}{\log l} = M \frac{n^3}{\log n}.$$

All the results in this paper are formulated for processes with discrete time, but they can be automatically reformulated for processes with continuous time by replacing certain sums by integrals.

The regularity conditions (I), (II) can, of course, be easily extended to the case of processes, made up of quantities which do not necessarily take numerical values. In particular, for vector processes, we can easily obtain analogues of the limit theorems proved in Sections 1 and 2, by using the methods developed by N. A. Sapogov [27].

1. Limit Theorems for Regular Processes

1.1. Properties of Regular Processes

1.1.1. *The regularity condition (I) is equivalent to the following condition: for any events $A \in \mathfrak{M}_{-\infty}^k$ and $B \in \mathfrak{M}_{k+n}^\infty$*

$$(1.1) \quad |\mathbf{P}(AB) - \mathbf{P}(A)\mathbf{P}(B)| \leq \varphi(n)\mathbf{P}(A).$$

PROOF. Suppose condition (I) is satisfied. Then

$$|\mathbf{P}(AB) - \mathbf{P}(A)\mathbf{P}(B)| = \left| \int_A (\mathbf{P}(B|\mathfrak{M}_{-\infty}^k) - \mathbf{P}(B)) \mathbf{P}(d\omega) \right| \leq \varphi(n)\mathbf{P}(A).$$

Next, we assume that condition (1.1) is satisfied and prove that condition (I) is also satisfied. In fact, if this were not the case, then we could find a number $n > 0$ and events $A \in \mathfrak{M}_{-\infty}^0$ and $B \in \mathfrak{M}_n^\infty$ for which $\mathbf{P}(A) > 0$ and

$$|\mathbf{P}(B|\mathfrak{M}_{-\infty}^0) - \mathbf{P}(B)| > \varphi(n),$$

if $\omega \in A$. Without loss of generality, we can assume that for $\omega \in A$

$$(1.2) \quad \mathbf{P}(B|\mathfrak{M}_{-\infty}^0) - \mathbf{P}(B) > \varphi(n).$$

Integrating both sides of (1.2) with respect to the event A , we arrive at a contradiction with (1.1).

1.1.2. Lemma 1.1. *Suppose the stationary process $\{x_j\}$ satisfies condition (I). If the random variable ξ is measurable with respect to $\mathfrak{M}_{-\infty}^k$ and the random variable η is measurable with respect to $\mathfrak{M}_{k+n}^\infty$, and if*

$$\mathbf{E}|\xi|^p < \infty, \quad \mathbf{E}|\eta|^q < \infty, \quad p, q > 1 \quad \text{and} \quad \frac{1}{p} + \frac{1}{q} = 1,$$

then

$$(1.3) \quad |\mathbf{E}\xi\eta - \mathbf{E}\xi\mathbf{E}\eta| \leq 2(\varphi(n))^{1/p} \mathbf{E}^{1/p}|\xi|^p \mathbf{E}^{1/q}|\eta|^q.$$

PROOF. We need only consider the case where

$$\begin{aligned} \xi &= \sum_j \lambda_j \chi(A_j), & A_{j_1} \cap A_{j_2} &= \emptyset, \quad j_1 \neq j_2, \\ \eta &= \sum_i \mu_i \chi(B_i), & B_{i_1} \cap B_{i_2} &= \emptyset, \quad i_1 \neq i_2. \end{aligned}$$

Here all the $A_j \in \mathfrak{M}_{-\infty}^k$, all the $B_i \in \mathfrak{M}_{k+n}^\infty$, and $\chi(A)$ denotes the characteristic function of the event A :

$$\chi(A) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

In our case

$$|\mathbf{E}\xi\eta - \mathbf{E}\xi\mathbf{E}\eta| = \left| \sum_{i,j} \lambda_j \mu_i \mathbf{P}(A_j B_i) - \sum_{i,j} \lambda_j \mu_i \mathbf{P}(A_j) \mathbf{P}(B_i) \right|.$$

Therefore, applying Hölder's inequality, we find that

$$\begin{aligned} |\mathbf{E}\xi\eta - \mathbf{E}\xi\mathbf{E}\eta| &= \left| \sum_j \lambda_j (\mathbf{P}(A_j))^{1/p} \cdot \sum_i [\mathbf{P}(B_i|A_j) - \mathbf{P}(B_i)] \mu_i (\mathbf{P}(A_j))^{1/q} \right| \\ &\leq \left(\sum_j |\lambda_j|^p \mathbf{P}(A_j) \right)^{1/p} \cdot \left(\sum_j \mathbf{P}(A_j) \left| \sum_i \mu_i [\mathbf{P}(B_i|A_j) - \mathbf{P}(B_i)] \right|^q \right)^{1/q} \\ &\leq \mathbf{E}^{1/p} |\xi|^p \sum_j \mathbf{P}(A_j) \left(\sum_i |\mu_i|^q (\mathbf{P}(B_i|A_j) + \mathbf{P}(B_i)) \right) \\ &\quad \times \left(\sum_i |\mathbf{P}(B_i|A_j) - \mathbf{P}(B_i)| \right)^{q/p|1/q}. \end{aligned}$$

Moreover, let A^+ denote the event $\mathbf{P}(B_i|A_j) - \mathbf{P}(B_i) > 0$, and let A^- denote the event $\mathbf{P}(B_i|A_j) - \mathbf{P}(B_i) < 0$. Then, taking account of condition (I), we have

$$\begin{aligned} \sum_i |\mathbf{P}(B_i|A_j) - \mathbf{P}(B_i)| &= (\mathbf{P}\{\bigcup_{A^+} B_i|A_j\} - \mathbf{P}\{\bigcup_{A^+} B_i\}) \\ &\quad + (\mathbf{P}\{\bigcup_{A^-} B_i\} - \mathbf{P}\{\bigcup_{A^-} B_i|A_j\}) \leq 2\varphi(n). \end{aligned}$$

From this we easily find that

$$|\mathbf{E}\xi\eta - \mathbf{E}\xi\mathbf{E}\eta| \leq 2(\varphi(n))^{1/p} \mathbf{E}^{1/p} |\xi|^p \mathbf{E}^{1/q} |\xi|^q.$$

1.1.3. Lemma 1.2. Suppose the stationary process $\{x_j\}$ satisfies condition (II). If the random variable ξ is measurable with respect to $\mathfrak{M}_\infty^k$ and the random variable η is measurable with respect to $\mathfrak{M}_{k+n}^\infty$, and if $|\xi| < C_1$, $|\eta| < C_2$, then

$$(1.4) \quad |\mathbf{E}\xi\eta - \mathbf{E}\xi\mathbf{E}\eta| \leq 4C_1 C_2 \alpha(n).$$

An inequality of the form (1.4), but with a constant equal to 16, is proved in [5]. Since the size of the constant will not be important in what follows, we omit the proof.

Lemma 1.3. (cf [25]). Suppose the stationary process $\{x_j\}$ satisfies the strong mixing condition (II). If the random variable ξ is measurable with respect to $\mathfrak{M}_\infty^k$ and the random variable η is measurable with respect to $\mathfrak{M}_{k+n}^\infty$, and if

$$\mathbf{E}|\xi|^{2+\delta} = C_1 < \infty \quad \text{and} \quad \mathbf{E}|\eta|^{2+\delta} = C_2 < \infty, \quad \delta > 0,$$

then

$$(1.5) \quad |\mathbf{E}\xi\eta - \mathbf{E}\xi\mathbf{E}\eta| \leq (4 + 2(C_1 + C_2 + \sqrt{C_1 C_2})) (\alpha(n))^{\delta/(2+\delta)}.$$

PROOF. From the condition of the lemma and the inequality (1.4), we have

$$\begin{aligned} |\mathbf{E}\xi\eta - \mathbf{E}\xi\mathbf{E}\eta| &\leq \left| \int_{\{|\xi| \leq N, |\eta| \leq N\}} \xi \eta \mathbf{P}(d\omega) - \int_{\{|\xi| \leq N\}} \xi \mathbf{P}(d\omega) \int_{\{|\eta| \leq N\}} \eta \mathbf{P}(d\omega) \right| \\ &\quad + \left| \int_{\{|\xi| \leq N, |\eta| > N\}} \xi \eta \mathbf{P}(d\omega) - \int_{\{|\xi| \leq N\}} \xi \mathbf{P}(d\omega) \int_{\{|\eta| > N\}} \eta \mathbf{P}(d\omega) \right| \end{aligned}$$

$$\begin{aligned}
& + \left| \int_{\{|\xi| > N, |\eta| \leq N\}} \xi \eta \mathbf{P}(d\omega) - \int_{\{|\xi| > N\}} \xi \mathbf{P}(d\omega) \int_{\{|\eta| \leq N\}} \eta \mathbf{P}(d\omega) \right| \\
& + \left| \int_{\{|\xi| > N, |\eta| > N\}} \xi \eta \mathbf{P}(d\omega) - \int_{\{|\xi| > N\}} \xi \mathbf{P}(d\omega) \int_{\{|\eta| > N\}} \eta \mathbf{P}(d\omega) \right| \\
& \leq 4N^2 \alpha(n) + \frac{2}{N^\delta} (C_1 + C_2) + \frac{2}{N^\delta} \sqrt{C_1 C_2}.
\end{aligned}$$

Setting

$$N = (\alpha(n))^{-1/(2+\delta)}$$

here, we obtain (1.5).

1.2. Possible Limit Distributions

Theorem 1.1. Suppose the stationary random process $\{x_i\}$ satisfies the strong mixing condition. Then the sequence of distribution functions

$$F_n(z) = \mathbf{P} \left\{ \frac{x_1 + x_2 + \cdots + x_n}{B_n} - A_n < z \right\},$$

where A_n and B_n are normalizing constants, can only converge to a stable law. If the limit law for $F_n(z)$ has exponent α , then $B_n = n^{1/\alpha} h(n)$, where $h(n)$ is a slowly varying function.²

PROOF. It follows from Lemma 1.2 that

$$\begin{aligned}
& \left| \mathbf{E} \exp \left\{ it \sum_{i=0}^r \left[\frac{1}{B_n} \sum_{j=1}^n x_{ik+in+j} - A_n \right] \right\} \right. \\
& \quad \left. - \prod_{i=0}^r \mathbf{E} \exp \left\{ it \left(\frac{1}{B_n} \sum_{j=1}^n x_{ik+in+j} - A_n \right) \right\} \right| \leq 4r\alpha(k).
\end{aligned}$$

From this inequality, just as in [22], for example, we easily deduce that the r -fold convolution of the function $F(z)$, which is the limit function for $F_n(z)$, is of the same type as $F(z)$ itself. The rest of the argument is the same as in [22], and the theorem is proved.

1.3. Variance of the Sums $S_n = x_1 + \cdots + x_n$

We now give some results concerning the behavior of the variance

$$\mathbf{E}(x_1 + \cdots + x_n)^2 = \sigma_n^2, \quad \mathbf{E}x_i = 0,$$

for large n , which will be needed later.

If $F(\lambda)$ denotes the spectral function of our process, then it is not hard to obtain the equality

$$(1.6) \quad \sigma_n^2 = \int_{-\pi}^{\pi} \frac{\sin^2 \frac{n\lambda}{2}}{\sin^2 \frac{\lambda}{2}} F(d\alpha).$$

² A function $h(n)$ is said to be slowly varying if

$$\lim_{n \rightarrow \infty} \frac{h(kn)}{h(n)} = 1$$

for any k .

Since the function

$$\frac{1}{n} \sin^2 \frac{n\lambda}{2} \bigg/ \sin^2 \frac{\lambda}{2}$$

differs from the Féjer kernel

$$\frac{1}{n\pi} \sin^2 \frac{n\lambda}{2} \bigg/ \sin^2 \frac{\lambda}{2}$$

only by a constant factor, it follows from (1.6) that

$$(1.7) \quad \sigma_n^2 = 2n\pi f(0) + o(n)$$

for an absolutely continuous spectral function $F(\lambda)$ with a spectral density $f(\lambda) = F'(\lambda)$ which is continuous at zero.

Below, in formulating certain limit theorems for sums of the form S_n , we shall require that $\sigma_n \rightarrow \infty$. In this regard, it is useful to note that the following alternative always holds for stationary processes with absolutely continuous spectral functions:

Theorem 1.2. *Let $\{x_j\}$ be a process which is stationary in the wide sense, and let $\sigma_n^2 = \mathbf{E}(x_1 + \cdots + x_n)^2$. Then, as $n \rightarrow \infty$, either $\sigma_n^2 \rightarrow \infty$ or $\sigma_n^2 < C < \infty$.*

PROOF. Since the spectral function of the process $\{x_j\}$ is absolutely continuous, it can be obtained by a process of moving averages (see [11], p. 499):

$$x_n = \sum_{j=-\infty}^{\infty} c_j \xi_{n+j},$$

where $\cdots, \xi_0, \xi_1, \cdots$ are mutually orthogonal random variables with $\mathbf{E}\xi_i\xi_j = \delta_{ij}$, and

$$\sum_{j=1}^{\infty} |c_j|^2 < \infty.$$

Therefore,

$$\sigma_n^2 = \sum |c_{i-1} + \cdots + c_{i-n}|^2.$$

We now assume that

$$\lim_{n \rightarrow \infty} \sigma_n^2 = C < \infty, \quad \overline{\lim}_{n \rightarrow \infty} \sigma_n^2 = \infty,$$

and we show that this assumption leads to a contradiction. Suppose the sequence $\{n_k\}$ is such that $\sigma_{n_k}^2 \rightarrow C$. It can be assumed that

[illegible]

But this and the following lemma imply that

$$(1.9) \quad \sum_{l=+1}^{+\infty} \left| A - \sum_{j=-1}^{-l} c_j \right|^2 \leq 2C < \infty.$$

Lemma 1.4. *If the numbers $a_l^{(r)} \geq 0$, $r, l = 1, 2, \dots$, if*

$$\sum_l a_l^{(r)} = \gamma(r) \leq C$$

and if

$$a_l^{(r)} \xrightarrow{r \rightarrow \infty} a_l,$$

then

$$\gamma = \sum_l a_l \leq C.$$

PROOF. According to the hypotheses of the lemma, we have

$$\left| \sum_{l=1}^R a_l - \sum_{l=1}^R a_l^{(r)} \right| \leq \sum_{l=1}^R |a_l - a_l^{(r)}|.$$

Since $|a_l - a_l^{(r)}| \rightarrow 0$, it follows that

$$\sum_{l=1}^R a_l \leq C \quad \text{and} \quad \sum_{l=1}^{\infty} a_l \leq C.$$

This proves the lemma and hence the inequality (1.9). From (1.9) it follows at once that

$$\sum_{j=-\infty}^{-1} c_j = A,$$

and moreover that

$$\sum_{j=0}^{\infty} c_j = -A,$$

since $\lim_{n \rightarrow \infty} \sigma_n^2 < \infty$.

Now let

$$c_{j-1} + c_{j-2} + \dots + c_{j-n} = c_{jn}.$$

Then

$$\sigma_n^2 = \sum_{-\infty}^{\infty} |c_{jn}|^2$$

and

$$\begin{aligned} c_{jn} &\rightarrow A - c_j - c_{j+1} - \dots - c_{-1} & \text{if } j < 0, \\ c_{jn} &\rightarrow A & \text{if } j = 0, \\ c_{jn} &\rightarrow A + c_0 + \dots + c_{j-n} & \text{if } j > 0, \end{aligned}$$

as $n \rightarrow \infty$. If we write

$$\lim_n c_{jn} = b_j,$$

then, since

$$\lim_{n_k} \sum_j |c_{jn_k}|^2 \leq C < \infty,$$

the lemma implies that

$$\sum |b_j|^2 \leq C,$$

as well. Thus, for all n we have

$$c_{jn} = c_{j-1} + \dots + c_{j-n} = b_j - b_{j+1-n}$$

and

$$\sigma_n^2 = \sum_j |c_{jn}|^2 \leq 2 \left(\sum_j |b_j|^2 + \sum_j |b_{j+1-n}|^2 \right) \leq 4C.$$

This completes the proof of Theorem 1.2.

Later on, we shall see that in many cases the variance σ_n^2 has the form

$$(1.10) \quad \sigma_n^2 = nh(n),$$

where $h(n)$ is a slowly varying function. We now formulate some lemmas exhibiting certain properties of the functions $h(n)$, which correspond to variances of sums of the values of a stationary process with an absolutely continuous spectral function.³

Lemma 1.5. *If k is fixed and $n \rightarrow \infty$, then*

$$\lim_n \frac{h(n+k)}{h(n)} = 1.$$

PROOF. We have

$$\begin{aligned} \sigma_{n+k}^2 &= \sigma_n^2 + \sigma_k^2 + 2\mathbf{E}(x_1 + \cdots + x_n)(x_{n+1} + \cdots + x_{n+k}), \\ |\mathbf{E}(x_1 + \cdots + x_n)(x_{n+1} + \cdots + x_{n+k})| &\leq \sigma_n \sigma_k. \end{aligned}$$

Moreover, $h(n)$ is a slowly varying function, and hence $nh(n) \rightarrow \infty$, $1/\sigma_n \rightarrow 0$. Therefore

$$\begin{aligned} \frac{\sigma_{n+k}^2}{\sigma_n^2} &= \frac{n+k}{n} \frac{h(n+k)}{h(n)} = 1 + o(1), \\ \frac{h(n+k)}{h(n)} &= 1 + o(1). \end{aligned}$$

Lemma 1.6. *As $s \rightarrow \infty$,*

$$\frac{1}{s} \sup_{2^s \leq p \leq 2^{s+1}} \left| \log \frac{h(p)}{h(2^s)} \right| \rightarrow 0.$$

PROOF. Let

$$\lambda(s) = \sup_{2^s \leq p \leq 2^{s+1}} \left| \log \frac{h(p)}{h(2^s)} \right|$$

and let the integers p_1 and p_2 be such that

$$\lambda(s) = \left| \log \frac{h(p_1)}{h(2^s)} \right| \quad \text{and} \quad \lambda(s+1) = \left| \log \frac{h(p_2)}{h(2^{s+1})} \right|.$$

Then

$$\lambda(s+1) \leq \lambda(s) + \left| \log \frac{h(2^s)}{h(2^{s+1})} \right| + \left| \log h \frac{h(p_2)}{\left(\left[\frac{p_2}{2}\right]\right)} \right| = \lambda(s) + \delta_1.$$

³ For an arbitrary slowly varying function with an integral-valued argument, these lemmas are not valid; however, for the case of a continuously varying argument, they are trivial consequences of a theorem of Karamata [17].

Similarly,

$$\lambda(s) \leq \lambda(s+1) + \left| \log \frac{h(2^{s+1})}{h(2^s)} \right| + \left| \log \frac{h(p_1)}{h(2p_1)} \right| = \lambda(s+1) + \delta_2.$$

From the definition of a slowly varying function and from Lemma 1.5, it follows that $\delta_1 \rightarrow 0$, $\delta_2 \rightarrow 0$ as $s \rightarrow \infty$. Therefore

$$\lambda(s+1) - \lambda(s) \rightarrow 0, \quad \lambda(s)/s \rightarrow 0.$$

Lemma 1.7. *If $c = c(n) \rightarrow 0$ as $n \rightarrow \infty$, then for any $\varepsilon > 0$,*

$$\lim_n \frac{h(cn)}{h(n)} c^\varepsilon = \lim_k \frac{h(n)}{h(cn)} c^\varepsilon = 0.$$

PROOF. We have

$$\begin{aligned} \log \frac{h(cn)}{h(n)} c^\varepsilon &= \varepsilon \log c + \sum_{k=0}^{k=\left[\frac{-\log c}{\log 2}\right]} \left(\log h\left(\left[\frac{n}{2^{k+1}}\right]\right) - \log h\left(\left[\frac{n}{2^k}\right]\right) \right) \\ (1.11) \quad &+ \log h(cn) - \log h\left(\left[\frac{n}{2^{\left[\frac{-\log c}{\log 2}\right]}}\right]\right). \end{aligned}$$

Because of Lemma 1.6,

$$\left| \log h(cn) - \log h\left(\left[\frac{n}{2^{\left[\frac{-\log c}{\log 2}\right]}}\right]\right) \right| < \frac{\varepsilon}{4} |\log c|,$$

for sufficiently large n . But, according to Lemma 1.5,

$$\left| \sum_k \left(\log h\left(\left[\frac{n}{2^{k+1}}\right]\right) - \log h\left(\left[\frac{n}{2^k}\right]\right) \right) \right| \leq \frac{\varepsilon}{4} |\log c|,$$

and therefore

$$\log \frac{h(cn)}{h(n)} c^\varepsilon < \frac{\varepsilon}{2} \log c \xrightarrow{n \rightarrow \infty} -\infty.$$

1.4. Necessary and Sufficient Conditions for Asymptotic Normality of the Sums $x_1 + x_2 + \cdots + x_n$

Theorem 1.3. *Suppose the stationary process $\{x_j\}$, where $\mathbf{E}x_j = 0$, $\mathbf{E}x_j^2 < \infty$, satisfies the strong mixing condition (II), and let*

$$S_n = \frac{x_1 + \cdots + x_n}{\sigma_n}, \quad \sigma_n^2 = \mathbf{E} \left(\sum_1^n x_j \right)^2.$$

Then a necessary and sufficient condition for the distribution functions of the sums S_n to converge to a normal law $\Phi(z)$ with parameters (0,1) is that the following conditions be satisfied:

1. $\sigma_n^2 = nh(n)$, where $h(n)$ is a slowly varying function.
2. For some $\varepsilon > 0$, the condition

$$\frac{k(n)}{\sigma_n^2} \int_{|z| > \varepsilon \sigma_n} z^2 dF_{[p(n)]}(z) \rightarrow 0$$

holds for any system of functions $k(n)$, $p(n)$ satisfying the requirements

- a) $k(n) \rightarrow \infty, p(n) \rightarrow \infty,$
- b) $\frac{(n-k(n)p(n))(k(n))^\beta}{p(n)} \rightarrow 0$ for some $\beta > 0,$
- c) $k(n)\alpha\left(\left[\frac{n-k(n)p(n)}{k(n)}\right]\right) \rightarrow 0.$

Here

$$F_i(z) = \mathbf{P}\{x_1 + \cdots + x_i < z\}.$$

PROOF. First we convince ourselves that systems of functions $p(n)$, $k(n)$ like those discussed in the theorem actually exist. Let $\lambda(n)$ be a function with the following properties:

$$\lambda(n) \rightarrow 0, \quad \frac{\lambda(n)}{\alpha([n^{1/4}])} \rightarrow \infty, \quad \lambda(n)n^{1/4} \rightarrow \infty.$$

(For example, we can set $\lambda(n) = \max\{\alpha([n^{1/4}])^{1/3}, 1/\log n\}$.) Now suppose that

(1.12)
$$k(n) = \min \left\{ \left[\frac{\lambda(n)}{\alpha([n^{1/4}])} \right], [\lambda(n)n^{1/4}] \right\},$$

(1.13)
$$p(n) = \left[\frac{n - kn^{1/4}}{k} \right].$$

Then it is not hard to see that the system of functions $p(n)$, $k(n)$ just constructed satisfies all the conditions of the theorem. In fact,

- a) $k(n) \rightarrow \infty, p(n) > \frac{n - \lambda(n)n^{1/2}}{\lambda(n)n^{1/4}} = \frac{n^{3/4}}{\lambda(n)} - n^{1/4} \rightarrow \infty,$
- b) $\frac{(n - k(n)p(n))(k(n))^\beta}{p(n)} \sim \frac{k^{1+\beta}n^{1/4}}{p(n)} < \frac{n^{1/2+\beta/4}}{n^{3/4}} \lambda^{1+\beta} \rightarrow 0,$
- c) $k(n)\alpha\left(\left[\frac{n - k(n)p(n)}{k(n)}\right]\right) \sim k(n)\alpha([n^{1/4}]) \leq \lambda(n) \rightarrow 0.$

We now turn to the proof of the theorem. Let $p(n)$, $k(n)$ be the system of functions of interest. We write

$$\left[\frac{n - k(n)p(n)}{k(n)} \right] = q(n) = q$$

and we introduce the notation

$$\begin{aligned} \xi_i &= \sum_{j=i p + i q + 1}^{(i+1)p + i q} x_j, & i &= 0, 1, \dots, k-1, \\ \eta_i &= \sum_{j=(i+1)p + i q + 1}^{(i+1)p + (i+1)q} x_j, & i &= 0, 1, \dots, k-1, \\ \eta_k &= \sum_{j=k p + k q + 1}^n x_j. \end{aligned}$$

The sum S_n can be represented in the form $S_n = S'_n + S''_n$, where

$$S'_n = \frac{1}{\sigma_n} \sum_{i=0}^{k-1} \xi_i, \quad S''_n = \frac{1}{\sigma_n} \sum_{i=0}^k \eta_i.$$

The sum S''_n converges in probability to zero. In fact,

$$\begin{aligned} \mathbf{E}|S''_n|^2 &= \frac{1}{\sigma_n^2} \sum_{i,j \leq k-1} \mathbf{E}\eta_i \eta_j + \frac{2}{\sigma_n^2} \sum_{i=0}^{k-1} \mathbf{E}\eta_i \eta_k + \frac{\mathbf{E}\eta_k^2}{\sigma_n^2} \\ (1.14) \quad &\leq \frac{k^2 q h(q)}{n h(n)} + \frac{2k \sqrt{q h(q) k h(k)}}{n h(n)} + \frac{k h(k)}{n h(n)}, \end{aligned}$$

since the number of terms in the expression for η_k equals

$$n - k(p+q) \leq n - \left(kp + k \frac{n - kp}{k} - k \right) = k.$$

All the terms in the right-hand side of the inequality (1.14) are infinitely small. According to Lemma 1.7,

$$\frac{k h(k)}{n h(n)} = \frac{k}{n} \frac{h\left(n \cdot \frac{k}{n}\right)}{h(n)} \rightarrow 0,$$

since

$$\frac{k}{n} = \frac{1}{p(n)} + \frac{o(1)}{nk^\beta} = o(1).$$

Again applying Lemma 1.7 and condition 1b of the theorem, we find that

$$\begin{aligned} \frac{k^2 q h(q)}{n h(n)} &\leq \frac{h(q)}{h(n)} \left(\frac{q}{n} \right)^{\beta_1} \left(\frac{q}{p} k^{2/(1-\beta_1)-1} \right)^{1-\beta_1} \\ &\sim \left(\frac{h\left(n \cdot \frac{q}{n}\right)}{h(n)} \left(\frac{q}{n} \right)^{\beta_1} \right) \left(\frac{n - kp}{k} k^{2\beta_1/(1-\beta_1)} \right)^{1-\beta_1} \xrightarrow{n \rightarrow \infty} 0, \end{aligned}$$

if $\beta_1 = \beta/(2+\beta)$, where β is the same as in condition 2b. Finally,

$$\frac{k \sqrt{q h(q) k h(k)}}{n h(n)} \leq \sqrt{\left(\frac{q}{n} \right)^{\beta_2} \frac{h(q)}{h(n)}} \sqrt{\left(\frac{k}{n} \right)^{\beta_2} \frac{h(k)}{h(n)}} \left(\frac{k^{\beta_2(1-\beta_2)} q}{p^2} \right)^{(1-\beta_2)/2} \rightarrow 0,$$

if $\beta_2 = \beta/(1+\beta)$, for example. Thus, $\mathbf{E}|S''_n|^2 \rightarrow 0$, $S''_n \rightarrow 0$, and therefore, the asymptotic behavior of the distribution functions of the random variables S_n is the same as that of the random variables S'_n . From (1.4) and condition 2b we find that

$$\left| \mathbf{E} e^{itS'_n} - \prod_{i=0}^{k-1} \mathbf{E} e^{it\xi_i/\sigma_n} \right| \leq 4k\alpha(q) \rightarrow 0.$$

Therefore, the distribution function of the sums

$$S'_n = \frac{1}{\sigma_n} \sum \xi_i$$

behaves in the limit as if the quantities ξ_i were independent. Moreover,

$$\mathbf{D}S'_n = \mathbf{D}S'_n - \mathbf{D}S''_n - 2\mathbf{E}S'_n S''_n = 1 + o(1).$$

Consequently, the quantities S'_n will be asymptotically normal if and only if the sums of the form

$$\frac{1}{\sigma_n} (\xi'_0 + \xi'_1 + \cdots + \xi'_{k-1})$$

are asymptotically normal, where the quantities ξ'_i are independent and distributed like ξ_i , i.e. (see [6], pp. 101–102) if

$$k \int_{|z| > \varepsilon} z^2 d_z \mathbf{P}\{\xi_0 < z\} = \frac{1}{\sigma^2} \int_{|z| > \varepsilon \sigma_n} z^2 dF_p(z) \rightarrow 0,$$

for any $\varepsilon > 0$. To complete the proof of the theorem, we have to verify the necessity of condition 1, but this follows at once from Theorem 1.1.

1.5. Sufficient Conditions for Asymptotic Normality

In this section, we show how to verify that the conditions of Theorem 1.3 are met, if we impose further restrictions on the moments of the quantities x_j and on the rate at which the functions $\varphi(n)$, $\alpha(n)$ fall off.

1.5.1. Theorem 1.4. *Suppose the stationary process $\{x_j\}$ satisfies condition (I), let $\mathbf{E}x_j = 0$, $\mathbf{E}|x_j|^{2+\delta} < \infty$ for some $\delta > 0$, and let $\sigma_n^2 \rightarrow \infty$, where σ_n^2 is the variance of the sum $x_1 + \cdots + x_n$. Then the random variables $S_n = (x_1 + \cdots + x_n)/\sigma_n$ are asymptotically normal.*

PROOF. It is sufficient to prove that conditions 1 and 2 of Theorem 1.3 are satisfied. We note that δ can be assumed to be less than 1.

Lemma 1.8. *The relation*

$$\sigma_n^2 = nh(n)$$

holds, where $h(n)$ is a slowly varying function.

PROOF. It is sufficient to show that

$$\lim_{n \rightarrow \infty} \sigma_{nk}^2 / \sigma_n^2 = k$$

for any k . We write

$$\begin{aligned} z_i &= x_{(i-1)n+(i-1)r+1} + \cdots + x_{in+(i-1)r}, & i &= 1, \cdots, k, \\ y_i &= x_{in+(i-1)r+1} + \cdots + x_{in+ir}, & i &= 1, 2, \cdots, k-1, \\ y_k &= -(x_{nk+1} + x_{nk+2} + \cdots + x_{n(k-1)r}), \end{aligned}$$

where $r = r(n) \rightarrow \infty$ as $n \rightarrow \infty$, but r increases so slowly that

$$\sigma_r / \sigma_n \rightarrow 0, \quad \sigma_{(k-1)r} / \sigma_n \rightarrow 0.$$

Then, taking account of (1.3), we find that

$$\begin{aligned} \sigma_{nk}^2 &= \mathbf{E}(z_1 + y_1 + z_2 + y_2 + \cdots + y_{k-1} + z_k + y_k)^2 \\ &= \sum_{i=1}^k \mathbf{E}z_i^2 + \sum_{i,j} \mathbf{E}z_i y_j + \sum_{i \neq j} \mathbf{E}z_i z_j + \sum_{i,j} \mathbf{E}y_i y_j = k\sigma_n^2(1 + o(1)), \end{aligned}$$

and the lemma is proved. The lemma immediately implies that condition 1 of Theorem 1.3 is met.

Lemma 1.9. *The inequality*

$$\mathbf{E}|x_1 + \cdots + x_n|^{2+\delta} \leq a|\sigma_n|^{2+\delta}$$

holds, where a is some constant.

PROOF (cf. [11], p. 225). Let

$$\mathbf{E}|x_1 + \cdots + x_n|^{2+\delta} = c_n.$$

Using (1.3), and arguing as in [11], pp. 225–226, we find that for any $\varepsilon > 0$ there exists a constant a_1 such that

$$c_{2n} \leq (2+\varepsilon)c_n + a_1\sigma_n^{2+\delta}, \quad n \geq 1.$$

Therefore, choosing ε sufficiently small, we find that for any $r \geq 1$

$$\begin{aligned} c_{2^r} &\leq (2+\varepsilon)^r c_1 + a_1 \{(\sigma_{2^{r-1}})^{2+\delta} + (2+\varepsilon)\sigma_{2^{r-2}}^{2+\delta} \\ &\quad \cdots + (2+\varepsilon)^{r-1}\} \leq (2+\varepsilon)^r c_1 + a_1(\sigma_{2^{r-1}})^{2+\delta} \gamma_r, \end{aligned}$$

where

$$\gamma_r = 1 + \frac{2+\varepsilon}{2^{1+\delta/2}} \left| \frac{h(2^{r-2})}{h(2^{r-1})} \right|^{1+\delta/2} + \cdots + \left(\frac{2+\varepsilon}{2^{1+\delta/2}} \right)^{r-1} \left| \frac{1}{h(2^{r-1})} \right|^{1+\delta/2}.$$

Next we show that $\gamma_r < a_2$ for sufficiently small ε , where a_2 is a constant. For any integer l in the interval $[2, r-1]$,

$$(1.15) \quad \frac{h(2^{r-l})}{h(2^{r-1})} = \left(\frac{h(2^{r-2})}{h(2^{r-1})} \frac{h(2^{r-3})}{h(2^{r-2})} \right) \cdots \frac{h(2^{r-s})}{h(2^{r-s+1})} \left(\frac{h(2^{r-s+1})}{h(2^{r-s})} \cdots \frac{h(2^{r-l})}{h(2^{r-l-1})} \right).$$

The function $h(n)$ is slowly varying, and hence for any $\varepsilon_1 > 0$ we can find an N such that

$$\frac{h(n)}{h(2n)} \leq 1 + \varepsilon_1$$

for $n > N$. Choosing s in formula (1.15) such that $2^{r-s} \geq N$, $2^{r-s-1} < N$, we have

$$\frac{h(2^{r-l})}{h(2^{r-1})} \leq (1+\varepsilon_1)^{s-1} \cdot a_3 \leq (1+\varepsilon_1)^{l-1} \cdot a_3,$$

where a_3 is a constant. If we now choose ε_1 and ε so small that

$$\frac{(1+\varepsilon_1)(2+\varepsilon)}{2^{1+\delta/2}} < \rho < 1,$$

we find that

$$\gamma_r = M/(1-\rho) < a_2.$$

Thus

(1.16)
$$\begin{aligned} c_{2^r} &\leq c_1(2+\varepsilon)^r + M(\sigma_2^r)^{2+\delta} \\ &= M(\sigma_{2^r})^{2+\delta} \left(1 + M \left(\frac{2+\varepsilon}{2^{1+\delta/2}} \right)^r \frac{1}{h(2^r)} \right) = M(\sigma_{2^r})^{2+\delta} \end{aligned}$$

for a suitable choice of ε .

Finally, let

$$2^r \leq n < 2^{r+1} \quad \text{and} \quad n = \nu_0 2^r + \nu_1 2^{r-1} + \cdots + \nu_r, \quad \nu_j = 0, 1, \quad j > 0; \quad \nu_0 = 1.$$

Using (1.11) and the Minkowski inequality, and representing the sum $x_1 + \cdots + x_n$ in the form

$$\begin{aligned} x_1 + \cdots + x_n &= (x_1 + \cdots + x_{i_1}) + (x_{i_1+1} + \cdots + x_{i_2}) \\ &\quad \cdots + (x_{i_{r+1}+1} + \cdots + x_{i_{r+1}}) \end{aligned}$$

(where the number of terms in the j -th set of parentheses equals $\nu_j 2^{r-j}$), we find that

$$\begin{aligned} c_n &\leq \left| \sum_{j=0}^r \mathbf{E}^{1/(2+\delta)} |x_1 + \cdots + x_{\nu_j 2^{r-j}}|^{2+\delta} \right|^{2+\delta} \leq M \left| \sum_{j=0}^r 2^{(r-j)/2} [h(2^{r-j})]^{1/2} \right|^{2+\delta} \\ &= M(h(n))^{1+\delta/2} \left(\frac{h(2^r)}{h(n)} \right)^{1+\delta/2} \left| \sum_{j=0}^r 2^{(r-j)/2} \left(\frac{h(2^{r-j})}{h(2^r)} \right)^{1/2} \right|^{2+\delta}. \end{aligned}$$

We now estimate the quantity

$$\frac{h(2^r)}{h(n)}, \quad \sum_{j=0}^r 2^{(r-j)/2} \left(\frac{h(2^{r-j})}{h(2^r)} \right)^{1/2}.$$

By an argument like that just given, we obtain

$$\begin{aligned} &2^{r/2} + 2^{(r-1)/2} \left(\frac{h(2^{r-1})}{h(2^r)} \right)^{1/2} + \cdots \\ &\leq 2^{r/2} \left(1 + \frac{1}{2} \left(\frac{h(2^{r-1})}{h(2^r)} \right)^{1/2} + \cdots \right) = M 2^{r/2} (1 + \rho_1 + \rho_1^2 + \cdots), \end{aligned}$$

where $0 < \rho_1 < 1$. Therefore

$$\left| \sum_{j=0}^r 2^{(r-j)/2} \left(\frac{h(2^{r-j})}{h(2^r)} \right)^{1/2} \right|^{2+\delta} = M 2^{r(2+\delta)/2} = M n^{1+\delta/2}.$$

Next, we prove that the ratio $h(2^r)/h(n)$ is bounded. Let $n - 2^r \geq (2^r + 2^{r+1})/2$ (the case $n - 2^r < (2^r + 2^{r+1})/2$ is handled similarly). We fix a value of m such that $\varphi(m) < 1/16$, and moreover we write

$$\begin{aligned} x_1 + x_2 + \cdots + x_n + x_{n+1} + \cdots + x_{n+m} &= (x_1 + \cdots + x_{2^r}) \\ &\quad + (x_{2^r+1} + \cdots + x_{2^r+m}) + (x_{2^r+m+1} + \cdots + x_{n+m}). \end{aligned}$$

From this we find that

$$(n+m)h(n+m) = 2^r h(2^r) + (n-2^r)h(n-2^r) + \theta,$$

where

$$\begin{aligned} |\theta| &\leq 2\sqrt{\varphi(m)}\sqrt{2^r(n-2^r)h(2^r)h(n-2^r)} \\ &\quad + 2\sqrt{2^r m h(2^r)h(m)} + 2\sqrt{(n-2^r)m h(m)h(n-2^r)} + m h(m). \end{aligned}$$

It is obvious that

$$2\sqrt{2^r(n-2^r)h(2^r)h(n-2^r)} \leq 2^r h(2^r) + (n-2^r)h(n-2^r).$$

Therefore, for large n ,

$$h(n+m) = \theta_1 h(2^r) + \theta_2 h(n-2^r) + r_n,$$

where

$$1 > \theta_1 > 1/4, \quad 1 > \theta_2 > 1/4, \quad r_n = o(n^{-1/4}),$$

and hence

$$1 = \theta_1 \frac{h(2^r)}{h(n+m)} + \theta_2 \frac{h(n-2^r)}{h(n+m)} + o(1),$$

since

$$n^{-1/4} h^{-1}(n+m) \rightarrow 0.$$

We now note that

$$h(n+m)/h(n) \rightarrow 1$$

for fixed m and $n \rightarrow \infty$. Therefore $h(2^r)/h(n) < 4$ for large n , and the lemma is proved.

It is now easy to finish the proof of the theorem. According to Theorem 1.3, we have to show that

$$\frac{k}{\sigma_n^2} \int_{|z| > \varepsilon \sigma_n} z^2 dF_p(z) \rightarrow 0.$$

The inequality of Lemma 1.9 gives

$$\frac{k}{\sigma_n^2} \int_{|z| > \varepsilon \sigma_n} z^2 dF_p(z) \leq \frac{k}{\varepsilon^\delta \sigma_n^{2+\delta}} \int |z|^{2+\delta} dF_p(z) \leq \frac{ak\sigma_p^{2+\delta}}{\varepsilon^\delta \sigma_n^{2+\delta}}.$$

Moreover, taking account of Lemma 1.7 and condition 2b of Theorem 1.3, we obtain

$$\frac{a}{\varepsilon^\delta} \frac{k\sigma_p^{2+\delta}}{\sigma_p^{2+\delta}} = \frac{a}{\varepsilon^\delta} \frac{k p^{1+\delta/2}}{n^{1+\delta/2}} \left(\frac{h(p)}{h(n)} \right)^{1+\delta/2} = \frac{a}{\varepsilon^\delta} \frac{p^{1+\delta/4} k}{n^{1+\delta/4}} \left(\frac{h(p)}{h(n)} \left(\frac{p}{n} \right)^{(\delta/4)2/(2+\delta)} \right)^{1+\delta/2} = o(1)$$

and the theorem is proved.

1.5.2. Theorem 1.5. *If the random process $\{x_j\}$ is stationary and satisfies condition (I), with*

$$\sum_n (\varphi(n))^{1/2} < \infty$$

and if $\mathbf{E}x_j = 0$, $\mathbf{E}x_j^2 < \infty$, then

$$(1.17) \quad \sigma^2 = \mathbf{E}x_0^2 + 2 \sum_{j=1}^{\infty} \mathbf{E}x_0 x_j < \infty.$$

Moreover, if $\sigma \neq 0$, then

$$\mathbf{P} \left(\frac{x_1 + \cdots + x_n}{\sigma \sqrt{n}} < z \right) \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

PROOF. Inequality (1.17) is an immediate consequence of the hypothesis of the theorem and the inequality

$$|R(j)| = |\mathbf{E}x_0x_j| \leq 2(\varphi(n))^{1/2}\mathbf{E}x_0^2,$$

which in turn is a consequence of (1.3). Therefore, under the conditions of the theorem,

$$\begin{aligned}\sigma_n^2 &= \mathbf{E}(x_1 + \cdots + x_n)^2 = n\mathbf{E}x_0^2 + 2 \sum_{j=1}^{n-1} (n-j)R(j) \\ &= n \left(R(0) + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) R(j) \right) = \sigma^2 n(1 + o(1)).\end{aligned}$$

Now let $\sigma \neq 0$. If $f(\cdot)$ is an arbitrary real function and if the process $\{x_j\}$ satisfies condition (I), then the process $\{f(x_j)\}$ also satisfies condition (I), at least with the same function $\varphi(n)$. In particular, the process $\{f_N(x_j)\}$, where

$$f_N(x) = \begin{cases} x, & |x| \leq N, \\ 0, & |x| > N, \end{cases}$$

satisfies condition (I). It is obvious that

$$\begin{aligned}\mathbf{E}|f_N(x_0)|^3 &< \infty, \\ \sigma_N^2 &= \mathbf{E}|f_N(x_0) - \mathbf{E}f_N(x_0)|^2 \\ &+ 2 \sum_{j=1}^{\infty} \mathbf{E}(f_N(x_0) - \mathbf{E}f_N(x_0))(f_N(x_j) - \mathbf{E}f_N(x_j)) < \infty\end{aligned}$$

and

$$\sigma_N^2(n) = \mathbf{E} \left| \sum_{j=1}^n (f_N(x_j) - \mathbf{E}f_N(x_j)) \right|^2 = n\sigma_N^2(1 + o(1)).$$

Therefore, according to Theorem 1.3,

$$\begin{aligned}(1.18) \quad \lim_n \mathbf{P} \left\{ \frac{f_N(x_1) + \cdots + f_N(x_n) - n\mathbf{E}f_N(x_0)}{\sigma_N(n)} < z \right\} \\ = \lim_n \mathbf{P} \left\{ \frac{\sum_{j=1}^n (f_N(x_j) - \mathbf{E}f_N(x_j))}{\sigma_N \sqrt{n}} < z \right\} = \Phi(z).\end{aligned}$$

Next we represent the sum

$$S_n = \frac{1}{\sigma \sqrt{n}} (x_1 + \cdots + x_n)$$

in the form $S_n = S'_n + S''_n$, where

$$\begin{aligned}S'_n &= \frac{\sum_{j=1}^n (f_N(x_j) - \mathbf{E}f_N(x_j))}{\sigma_N \sqrt{n}} \cdot \frac{\sigma_N}{\sigma_n}, \\ S''_n &= \frac{\sum_{j=1}^n (\bar{f}_N(x_j) - \mathbf{E}\bar{f}_N(x_j))}{\sigma \sqrt{n}},\end{aligned}$$

and

$$\bar{f}_N(x) = x - f_N(x) = \begin{cases} 0, & |x| \leq N, \\ x, & |x| > N. \end{cases}$$

Evaluating $\mathbf{E}|S_n''|^2$, we have

$$\begin{aligned} \mathbf{E}|S_n''|^2 &= \frac{1}{\sigma^2 n} (n\mathbf{E}|\bar{f}_N(x_1) - \mathbf{E}\bar{f}_N(x_1)|^2 \\ &\quad + 2 \sum_{j=1}^{n-1} (n-j)\mathbf{E}(\bar{f}_N(x_0) - \mathbf{E}\bar{f}_N(x_0))(\bar{f}_N(x_j) - \mathbf{E}\bar{f}_N(x_j))). \end{aligned}$$

Because of (1.3),

$$\begin{aligned} |\mathbf{E}(\bar{f}_N(x_0) - \mathbf{E}\bar{f}_N(x_0))(\bar{f}_N(x_j) - \mathbf{E}\bar{f}_N(x_j))| &\leq 2(\varphi(j))^{1/2} \mathbf{E}|\bar{f}_N(x_0) - \mathbf{E}\bar{f}_N(x_0)|^2 \\ &\leq 4(\varphi(j))^{1/2} \mathbf{E}|\bar{f}_N(x_0)|^2 = r_N(\varphi(j))^{1/2} \end{aligned}$$

for all $j = 0, 1, \dots$, if we set $\varphi(0) = 1$, $r_N = 4\mathbf{E}|\bar{f}_N(x_0)|^2$, and therefore

$$\mathbf{E}|S_n''|^2 \leq \frac{r_N}{\sigma^2} \left(1 + 2 \sum_{j=1}^{\infty} (\varphi(j))^{1/2} \right).$$

Moreover, for any $\varepsilon > 0$ whatsoever, we can always choose N so large that

$$\mathbf{E}|S_n''|^2 \leq \varepsilon \quad \text{and} \quad \left| 1 - \frac{\sigma_N}{\sigma} \right| \leq \varepsilon.$$

If now $f_n(t) = \mathbf{E}e^{itS_n}$ is the characteristic function of the quantity S_n , then

$$\begin{aligned} (1.19) \quad |f_n(t) - e^{-t^2/2}| &\leq |e^{-(t^2/2)(\sigma_N^2/\sigma^2)} - e^{-t^2/2}| + |e^{-(t^2/2)(\sigma_N^2/\sigma^2)} - \mathbf{E}e^{itS_n'}| + |t|\mathbf{E}|S_n''| \\ &\leq |t|\sqrt{\varepsilon} + \varepsilon t^2 + |e^{-(t^2/2)(\sigma_N^2/\sigma^2)} - \mathbf{E}e^{itS_n'}|. \end{aligned}$$

Because of (1.18), the last expression in (1.19) converges to zero as $n \rightarrow \infty$, and the theorem is proved.

1.5.3. Theorem 1.6. *If the random process $\{x_j\}$ is stationary and satisfies the strong mixing condition (II), with*

$$\sum_1^{\infty} \alpha(n) < \infty \quad \text{and} \quad \alpha(n) < \frac{M}{n \log n},$$

and if $\mathbf{E}x_j = 0$ and the random variables x_j are bounded, i.e.,

$$|x_j| < C < \infty$$

with probability 1, then

$$(1.20) \quad \sigma^2 = \mathbf{E}x_0^2 + 2 \sum_{j=1}^{\infty} \mathbf{E}x_0 x_j < \infty.$$

Moreover, if $\sigma \neq 0$, then

$$\mathbf{P} \left\{ \frac{x_1 + \dots + x_n}{\sigma\sqrt{n}} < z \right\} \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

PROOF. The convergence of the series (1.20) is easily deduced from the inequality $|\mathbf{E}x_0 x_j| \leq 4C^2 \alpha(n)$ and the conditions of the theorem. From (1.20),

just as in the proof of Theorem 1.5, we deduce that

$$(1.21) \quad \sigma_n^2 = \mathbf{E} \left(\sum_{j=1}^n x_j \right)^2 = \sigma^2 \cdot n(1+o(1))$$

and hence that

$$\lim_n \mathbf{P} \left(\frac{x_1 + \cdots + x_n}{\sigma_n} < z \right) = \lim_n \mathbf{P} \left(\frac{x_1 + \cdots + x_n}{\sigma \sqrt{n}} < z \right)$$

for all z , provided only that the limit in the left-hand side exists. We now estimate the growth of the fourth moment

$$\mathbf{E} \left(\sum_{j=1}^n x_j \right)^4.$$

We have

$$(1.22) \quad \mathbf{E} \left(\sum_1^n x_j \right)^4 = n \mathbf{E} x_1^4 + \sum_{i \neq j} \mathbf{E} x_i^2 x_j^2 + \sum_{i \neq j} \mathbf{E} x_i^3 x_j + \sum_{i \neq j \neq k} x_i^2 x_j x_k + \sum_{i \neq j \neq k \neq m} \mathbf{E} x_i x_j x_k x_m.$$

Using the inequality (1.4), we easily deduce from the conditions of the theorem that

$$\left| \sum_{i \neq j \neq k} \mathbf{E} x_i^2 x_j x_k \right| = MC^4 n^2 \log n, \quad \left| \sum_{i \neq j \neq k \neq m} \mathbf{E} x_i x_j x_k x_m \right| = MC^4 \frac{n^3}{\log n}$$

and hence that

$$\mathbf{E} \left(\sum_{j=1}^n x_j \right)^4 = M \frac{n^3}{\log n}.$$

Just as before (see the proof of Theorem 1.3), we introduce the random variables $\xi_i, \eta_i \eta_k, i = 0, \dots, k-1$, and we define the functions $p(n), q(n)$ and $k(n)$ as follows:

$$p(n) = [n^{1/2} \log^{3/8} n], \quad q(n) = [n^{1/2} \log^{-3/8} n], \\ k(n) = \left[\frac{n}{p+q} \right] \sim n^{1/2} \log^{-3/8} n.$$

Bearing in mind that $|j-i| = s, |\mathbf{E} \eta_i \eta_j| \leq 4q^2 C^2 \alpha(ps)$ and that

$$\mathbf{E} \eta_k^2 = M \sigma^2 (p+q), \\ |\mathbf{E} \eta_i \eta_k| \leq 4C^2 q(p+q) \alpha((k-i)p), \quad i = 0, 1, \dots, k-1,$$

we easily obtain the inequality

$$\mathbf{E} |S_n''| \leq M \log^{-1/2} n.$$

Moreover

$$\left| \mathbf{E} e^{it(\xi_0 + \cdots + \xi_{k-1})} - \prod_{i=0}^{k-1} \mathbf{E} e^{i\xi_i t} \right| \leq 4k\alpha(q) = M \log^{-1} n,$$

and finally

$$\frac{k}{\sigma^2 n} \int_{|x_1 + \cdots + x_p| > \varepsilon \sqrt{n}} |x_1 + \cdots + x_p|^2 P(d\omega) = \frac{Mkp^2}{\varepsilon^3 n^2 \log p} = \frac{M}{\varepsilon} \log^{-1/4} n.$$

approximated by functions of a finite number of the x_j , then we can expect that the process $\{f_j\}$ will closely resemble a regular process, and, in particular, that it will obey the central limit theorem, under sufficiently broad conditions. These conditions will be given in the theorems formulated below.

One further fact should be noted. If the process $\{x_j\}$ satisfies conditions (I) and (II), and if the stationary process $\{f_j\}$ is generated by the process $\{x_j\}$ in the way just described, then the spectral function of the process $\{f_j\}$ is absolutely continuous (see e.g., [4]).

2.1.2. Theorem 2.1. *Let the stationary process $\{x_j\}$ satisfy condition (I), let the random variable f be measurable with respect to $\mathfrak{M}_{-\infty}^\infty$, and let the process $\{f_i\}$ be obtained from f , as described above. Moreover, suppose the following conditions are satisfied:*

- 1. $\mathbf{E}f = 0, \quad \mathbf{E}f^2 < \infty,$
- 2. $\sum_{k=1}^\infty |\mathbf{E}\{ |f - \mathbf{E}\{f|\mathfrak{M}_{-k}^k\}|^2 \}|^{1/2} < \infty,$
- 3. $\sum_{j=1}^\infty (\varphi(k))^{1/2} < \infty.$

Then

(2.2)
$$\sigma^2 = \mathbf{E}f_0^2 + 2 \sum_{j=1}^\infty \mathbf{E}\{f_0 f_j\} < \infty;$$

and moreover

$$\mathbf{P}\left\{\frac{f_1 + \cdots + f_n}{\sigma\sqrt{n}} < z\right\} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du$$

if $\sigma \neq 0$.

PROOF. If $\xi_j^{(s)} = \mathbf{E}\{f_j|\mathfrak{M}_{-s+j}^{s+j}\}$, the stationary process $\{\xi_j^{(s)}\}$ satisfies condition (I). In fact, if $\mathfrak{M}_a^b(\xi)$ are the σ -algebras generated by the process $\{\xi_j^{(s)}\}$, then $\mathfrak{M}_{k+n}^\infty(\xi) \subset \mathfrak{M}_{k+n-s}^\infty$ and $\mathfrak{M}_{-\infty}^k(\xi) \subset \mathfrak{M}_{-\infty}^{k+s}$, and therefore for $A \in \mathfrak{M}_{-\infty}^k(\xi)$ and $B \in \mathfrak{M}_{k+n}^\infty(\xi)$ we have the inequality

$$|\mathbf{P}(AB) - \mathbf{P}(A)\mathbf{P}(B)| \leq \varphi_\xi(n)\mathbf{P}(A),$$

where

(2.3)
$$\varphi_\xi(n) \leq \begin{cases} 1, & n \leq 2s \\ \varphi(n-2s), & n > 2s. \end{cases}$$

Moreover, since (see [11], Chap. 1, Sec. 10)

$$\mathbf{E}\{|\xi_j^{(s)}|^2\} = \mathbf{E}\{\mathbf{E}^2\{f_j|\mathfrak{M}_{-s+j}^{s+j}\}\} \leq \mathbf{E}\{\mathbf{E}\{f_j^2|\mathfrak{M}_{-s+j}^{s+j}\}\} = \mathbf{E}f^2 < \infty,$$

the stationary process $\{\xi_j^{(s)}\}$ satisfies all the conditions of Theorem 1.4, and hence

(2.4)
$$\mathbf{P}\left\{\frac{\xi_1^{(s)} + \xi_2^{(s)} + \cdots + \xi_n^{(s)}}{\sigma_s\sqrt{n}} < z\right\} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du$$

if

$$\sigma_s^2 = \mathbf{E}|\xi_0^{(s)}|^2 + 2 \sum_{j=1}^\infty \mathbf{E}\{\xi_0^{(s)} \xi_j^{(s)}\} \neq 0.$$

Next we set $\eta_j^{(s)} = f_j - \xi_j^{(s)}$ and estimate the correlation functions of the processes $\{f_j\}$ and $\{\eta_j^{(s)}\}$. Setting

$$\mathbf{E}|f - \mathbf{E}\{f|\mathfrak{M}_{-k}^k\}|^2 = \psi(k)$$

and taking account of (3.3), we have

$$\begin{aligned} \mathbf{E}f_0f_j &= \mathbf{E}(\xi_0^{([j/3])} + \eta_0^{([j/3])})(\xi_j^{([j/3])} + \eta_j^{([j/3])}) \\ (2.5) \quad &\leq 2\varphi^{1/2}\left(\left[\frac{j}{3}\right]\right)\mathbf{E}f^2 + 2\mathbf{E}^{1/2}f^2\left(\varphi^{1/2}\left(\left[\frac{j}{3}\right]\right) + \psi\left(\left[\frac{j}{3}\right]\right)\right) \\ &= M\left(\varphi^{1/2}\left(\left[\frac{j}{3}\right]\right) + \psi^{1/2}\left(\left[\frac{j}{3}\right]\right)\right). \end{aligned}$$

The process $\{\eta_j^{(s)}\}$ is constructed in the same way as the process $\{f_j\}$, i.e.,

$$\begin{aligned} \eta_j^{(s)} &= T_1^j\eta_0^{(s)} = T_1^j\eta^{(s)}(\cdots x_{-1}, x_0, x_1, \cdots), \\ \mathbf{E}\{|\eta_0^{(s)} - \mathbf{E}\{\eta_0^{(s)}|\mathfrak{M}_{-k}^k\}|^2\} &= \mathbf{E}\{f - \mathbf{E}\{f|\mathfrak{M}_{-s}^s\} - \mathbf{E}\{f|\mathfrak{M}_{-k}^k\} + \mathbf{E}\{\mathbf{E}\{f|\mathfrak{M}_{-s}^s\}|\mathfrak{M}_{-k}^k\}|^2\} \\ &= \psi_s(k) = \begin{cases} \psi(s), & k \leq s \\ \psi(k), & k > s, \end{cases} \end{aligned}$$

since

$$\mathbf{E}\{\mathbf{E}\{f|\mathfrak{M}_{-s}^s\}|\mathfrak{M}_{-k}^k\} = \begin{cases} \mathbf{E}\{f|\mathfrak{M}_{-s}^s\}, & k \geq s \\ \mathbf{E}\{f|\mathfrak{M}_{-k}^k\}, & k \leq s, \end{cases}$$

(cf. [11], Chap. 1, Secs. 7–10). Therefore, by the same argument as used to estimate (2.5), we find that

$$(2.6) \quad |\mathbf{E}\eta_0^{(s)}\eta_j^{(s)}| = M\left(\varphi^{1/2}\left(\left[\frac{j}{3}\right]\right) + \psi_s^{1/2}\left(\left[\frac{j}{3}\right]\right)\right) = M\left(\varphi^{1/2}\left(\left[\frac{j}{3}\right]\right) + \psi^{1/2}\left(\left[\frac{j}{3}\right]\right)\right).$$

The convergence of the series (2.2) is now an immediate consequence of conditions 2 and 3 of the theorem and the estimate (2.5).

We now represent the sum

$$S_n = \frac{1}{\sigma\sqrt{n}} \sum_{j=1}^n f_j$$

in the form $S_n = S'_n + S''_n$, where

$$S'_n = \frac{1}{\sigma\sqrt{n}} \sum_{j=1}^n \xi_j^{(s)}, \quad S''_n = \frac{1}{\sigma\sqrt{n}} \sum_{j=1}^n \eta_j^{(s)}.$$

Taking account of (2.6) we find that

$$\begin{aligned} \mathbf{E}|S''_n|^2 &= \frac{1}{\sigma^2 n} \left(n\mathbf{E}|\eta_0^{(s)}|^2 + 2 \sum_{j=1}^{n-1} (n-j)\mathbf{E}\eta_0^{(s)}\eta_j^{(s)} \right) \\ &\leq (2N+1)\psi(s) + M \sum_{j=N}^n \left(\varphi^{1/2}\left(\left[\frac{j}{3}\right]\right) + \psi^{1/2}\left(\left[\frac{j}{3}\right]\right) \right). \end{aligned}$$

Choosing first N and then s sufficiently large, we can make $\mathbf{E}|S''_n|^2$ as small as we please. Moreover, for large s we can also make the expression $|1 - (\sigma_s/\sigma)|$ small. Therefore, choosing sufficiently large N and s , and then letting n go to

infinity, we can make the difference

$$\mathbf{E}e^{itS_n} - e^{-t^2/2} \leq |t| \mathbf{E}|S_n| S_n'' + |e^{-(t^2/2)(\sigma_s^2/\sigma^2)} - e^{-t^2/2}| + |e^{-(t^2/2)(\sigma_s^2/\sigma^2)} - \mathbf{E}e^{itS_n}|$$

as small as we please. This proves the theorem.

If the random variable f has moments of order higher than 2, we can weaken the conditions of the theorem somewhat.

2.1.3. Theorem 2.2. *Let the stationary process $\{x_j\}$ satisfy the strong mixing condition (II), let the random variable f be measurable with respect to $\mathfrak{M}_{-\infty}^\infty$, and let the process $\{f_j\}$ be obtained from f in the same way as in Theorem 2.1. Moreover, suppose the following conditions are satisfied:*

1. $\mathbf{E}f = 0$ and for some $\delta > 0$, $\mathbf{E}|f|^{2+\delta} < \infty$,
2. $\sum_{k=1}^{\infty} |\mathbf{E}\{f - \mathbf{E}\{f | \mathfrak{M}_{-k}^k\}\}^{(2+\delta)/(1+\delta)}|^{(1+\delta)/(2+\delta)} < \infty$,
3. $\sum_{k=1}^{\infty} (\alpha(k))^{\delta/(2+\delta)} < \infty$.

Then

$$(2.7) \quad \sigma^2 = \mathbf{E}f_0^2 + 2 \sum_{j=1}^{\infty} \mathbf{E}f_0 f_j < \infty,$$

and moreover

$$(2.8) \quad \mathbf{P}\left\{\frac{f_1 + f_2 + \cdots + f_n}{\sigma\sqrt{n}} < z\right\} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du$$

if $\sigma \neq 0$.

PROOF. Let $\xi_j^{(s)}$ and $\eta_j^{(s)}$ be the same as before. According to condition 1 of the theorem,

$$(2.9) \quad \mathbf{E}|\xi_j^{(s)}|^{2+\delta} = \mathbf{E}\{|\mathbf{E}\{f_j | \mathfrak{M}_{-s+j}^{s+j}\}|^{2+\delta}\} \leq \mathbf{E}\{|\mathbf{E}\{f_j | \mathfrak{M}_{-s+j}^{s+j}\}|^{2+\delta}\} = \mathbf{E}|f|^{2+\delta} < \infty,$$

and, according to the Minkowski inequality,

$$(2.10) \quad \mathbf{E}|\eta_j^{(s)}|^{2+\delta} = \mathbf{E}|f_j - \xi_j^{(s)}|^{2+\delta} \leq 2^{2+\delta} \mathbf{E}|f|^{2+\delta}.$$

From (2.9) and Lemma 1.3, we find that

$$(2.11) \quad |\mathbf{E}\xi_0^{(s)} \xi_j^{(s)}| = M(\alpha(j-2s))^{\delta/(2+\delta)}.$$

Moreover, using Hölder's inequality, we find that

$$(2.12) \quad |\mathbf{E}\xi_i^{(s)} \eta_j^{(s)}| \leq \mathbf{E}^{(1+\delta)/(2+\delta)}\{|\eta_i^{(s)}|^{(2+\delta)/(1+\delta)}\} \mathbf{E}^{1/(2+\delta)}\{|\xi_j^{(s)}|^{2+\delta}\}$$

and

$$(2.13) \quad |\mathbf{E}\eta_i^{(s)} \eta_j^{(s)}| \leq \mathbf{E}^{(1+\delta)/(2+\delta)}\{|\eta_i^{(s)}|^{(2+\delta)/(1+\delta)}\} \mathbf{E}^{1/(2+\delta)}\{|\eta_j^{(s)}|^{2+\delta}\}.$$

From (2.11), (2.12) and (2.13), as in the case of (2.5) and (2.6), we obtain

$$\begin{aligned} |\mathbf{E}f_0 f_j| &= M\left(\left(\alpha\left(\left[\frac{j}{3}\right]\right)\right)^{\delta/(2+\delta)} + \mathbf{E}^{(1+\delta)/(2+\delta)}|\eta_0^{([j/3])}|^{(2+\delta)/(1+\delta)}\right), \\ |\mathbf{E}\eta_0^{(s)} \eta_j^{(s)}| &= M\left(\left(\alpha\left(\left[\frac{j}{3}\right]\right)\right)^{\delta/(2+\delta)} + \mathbf{E}^{(1+\delta)/(2+\delta)}|\eta_0^{([j/3])}|^{(2+\delta)/(1+\delta)}\right), \\ |\mathbf{E}\eta_0^{(s)} \eta_j^{(s)}| &\leq \mathbf{E}|\eta_0^{(s)}|^2. \end{aligned}$$

The rest of the argument is the same as in Theorem 2.1.

REMARK. Condition 2 of Theorem 2.2 is weaker than condition 2 of Theorem 2.1, since for $\delta > 0$

$$\mathbf{E}^{(1+\delta)/(2+\delta)}|f - \mathbf{E}\{f|\mathfrak{M}_{-k}^k\}|^{(2+\delta)/(1+\delta)} \leq \mathbf{E}^{1/2}|f - \mathbf{E}\{f|\mathfrak{M}_{-k}^k\}|^2.$$

Theorem 2.3. *Formulas (2.7) and (2.8) remain valid if the conditions 1, 2 and 3 of Theorem 2.2 are replaced by the following conditions:*

1. $|f| < C < \infty$, $\mathbf{E}f = 0$,
2. $\sum_{k=1}^{\infty} \mathbf{E}|f - \mathbf{E}\{f|\mathfrak{M}_{-k}^k\}| < \infty$,
3. $\sum_{k=1}^{\infty} \alpha(k) < \infty$ and $\alpha(k) \leq \frac{M}{k \log k}$.

The proof is similar to that of Theorem 2.2; we have to use Theorem 1.5 and the inequalities

$$\begin{aligned} |\mathbf{E}\xi_0^{(s)}\xi_j^{(s)}| &\leq 4C^2\alpha(j-2s), \\ |\mathbf{E}\xi_i^{(s)}\eta_j^{(s)}| &\leq C\mathbf{E}|\eta_j^{(s)}|, \quad |\mathbf{E}\eta_i^{(s)}\eta_j^{(s)}| \leq \mathbf{E}|\eta_0^{(s)}|^2, \\ |\mathbf{E}f_0f_j| &= M\left(\alpha\left(\left[\frac{j}{3}\right]\right) + \mathbf{E}(|\eta_0^{([j/3])}|)\right). \end{aligned}$$

REMARK. It is easily verified that all the theorems proved above remain valid for a definition of the process $\{f_j\}$ which is broader in a formal sense (and is more convenient in many cases). Let $\mathfrak{M}_a^b, M_a^b \subset \mathfrak{M}_a^{b'}$ be a family of σ -algebras of events, which satisfies the strong mixing condition (or condition (I)). Moreover, suppose a transformation T which preserves the probability of events is defined on $\mathfrak{M}_{-\infty}^{\infty}$, with $T\mathfrak{M}_a^b = \mathfrak{M}_{a+1}^{b+1}$, and let T_1 denote the transformation of the random variables measurable with respect to $\mathfrak{M}_{-\infty}^{\infty}$ which is induced by the transformation T . In this case, the process $\{f_j\}$ of interest to us is defined by the relations

$$f_j = T_1^j f, \quad -\infty < j < \infty,$$

where the random variable f is measurable with respect to $\mathfrak{M}_{-\infty}^{\infty}$.

2.1.4. We now give two more theorems concerning stationary processes which are obtained by a process of moving averages from processes obeying the central limit theorem.

Theorem 2.4.⁴ *Let $\{x_j\}$ be a process which is stationary in the wide sense (and not necessarily regular), with $\mathbf{E}x_0 = 0$ and $\mathbf{E}x_0^2 < \infty$, and let*

$$\sigma_n^2 = \mathbf{E}(x_1 + \cdots + x_n)^2 \xrightarrow{n \rightarrow \infty} \infty.$$

Suppose

$$y_j = \sum_{-\infty}^{\infty} c_{k-j} x_k, \quad \text{where} \quad \sum_{-\infty}^{\infty} |c_k| < \infty.$$

Then, if

⁴ Cf. [7], [21].

$$(2.14) \quad \mathbf{P} \left\{ \frac{x_1 + \cdots + x_n}{\sigma_n} < z \right\} \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du$$

as $n \rightarrow \infty$, we have

$$(2.15) \quad \mathbf{P} \left\{ \frac{y_1 + \cdots + y_n}{\sigma_n} < z \right\} \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z/\sigma} e^{-u^2/2} du,$$

where

$$\sigma = \left| \sum_{-\infty}^{\infty} c_k \right|.$$

Conversely, if in addition $\sigma \neq 0$, then (2.15) implies (2.14).

PROOF. Setting

$$y_{0N} = \sum_{-N}^N c_j x_j, \quad y_{kN} = T_1^k y_{0N} = \sum_{-N}^N c_j x_{j+k},$$

we show that the sums

$$S'_n = \frac{1}{\sigma_n} (y_{1N} + \cdots + y_{nN})$$

are asymptotically normal. In fact

$$(2.16) \quad \frac{1}{\sigma_n} \sum_{k=1}^n y_{kN} = \frac{1}{\sigma_n} \sum_{k=1}^n \sum_{j=-N}^N c_j x_{j+k} = \frac{1}{\sigma_n} \sum_{j=-N}^N c_j \left(\sum_{k=1}^n x_k + R_{nN} \right),$$

where

$$|R_{nN}| \leq \frac{1}{\sigma_n} \sum_{j=-N}^N |c_j| \left(\sum_{i=-N}^{-1} |x_i| + \sum_{i=n+1}^{N+n} |x_i| \right).$$

For fixed N and $n \rightarrow \infty$, $R_{nN} \rightarrow 0$, and therefore

$$(2.17) \quad \lim_n \mathbf{P} \left\{ \frac{1}{\sigma_n} \sum_{k=1}^n y_{kN} < z \right\} = \lim_n \mathbf{P} \left\{ \frac{1}{\sigma_n} \sum_{j=-N}^N c_j \sum_{k=1}^n x_k < z \right\} \\ = \frac{1}{2\pi} \int_{-\infty}^{z(\sum_{j=-N}^N c_j)^{-1}} e^{-u^2/2} du.$$

We now write the sum

$$S_n = \frac{1}{\sigma_n} (y_1 + \cdots + y_n)$$

in the form $S_n = S'_n + S''_n$, where

$$S''_n = \frac{1}{\sigma_n} (\tilde{y}_{1N} + \cdots + \tilde{y}_{nN}), \\ \tilde{y}_{iN} = y_i - y_{iN}.$$

Because of (2.17), it is sufficient to show that $\mathbf{E}|S''_n|^2 \leq \delta(N)$, where $\delta(N) \rightarrow 0$ as $N \rightarrow \infty$. Turning to the spectral representation of the process $\{x_j\}$, we have

$$x_j = \int_{-\pi}^{\pi} e^{i\lambda j} Z(d\lambda),$$

where $Z(\Delta)$ is an orthogonal random measure. Then, as is well known ([11], Chap. 10), the spectral representation of the process $\{\bar{y}_{iN}\}$ can be written in the form

$$\bar{y}_{jN} = \int_{-\pi}^{\pi} e^{i\lambda j} c_N(\lambda) Z(d\lambda),$$

where

$$c_N(\lambda) = \sum_{|j| > N} c_j e^{i\lambda j}.$$

Therefore, the variances σ_n^2 and $\mathbf{E}|S_n''|^2$ can be expressed in terms of the spectral function $F(\lambda)$ of the process $\{x_j\}$ in the following way:

$$(2.18) \quad \sigma_n^2 = \int_{-\pi}^{\pi} \frac{\sin^2 \frac{n\lambda}{2}}{\sin^2 \frac{\lambda}{2}} F(d\lambda),$$

$$(2.19) \quad \mathbf{E}|S_n''|^2 = \frac{1}{\sigma_n^2} \int_{-\pi}^{\pi} \frac{\sin^2 \frac{n\lambda}{2}}{\sin^2 \frac{\lambda}{2}} |c_N(\lambda)|^2 F(d\lambda).$$

Since (2.18) and (2.19) imply

$$\mathbf{E}|S_n''|^2 \leq \frac{\max_{\lambda \in [-\pi, \pi]} |c_N(\lambda)|^2}{\sigma_n^2} \int_{-\pi}^{\pi} \frac{\sin^2 \frac{n\lambda}{2}}{\sin^2 \frac{\lambda}{2}} F(d\lambda) \leq \left(\sum_{|j| > N} |c_j|^2 \right) \xrightarrow[N \rightarrow \infty]{} 0,$$

the theorem is proved.

Theorem 2.5.⁵ Let $\{x_j\}$ be a sequence of independent, identically distributed random variables with $\mathbf{E}x_0 = 0$, $\mathbf{E}x_j^2 < \infty$, and let

$$y_j = \sum_{k=-\infty}^{\infty} c_{k-j} x_k, \quad \text{where} \quad \sum_{k=-\infty}^{\infty} |c_k|^2 < \infty.$$

Then

$$\mathbf{P} \left\{ \frac{y_1 + \cdots + y_n}{\sigma_n} < z \right\} \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du$$

for all z , if $\sigma_n^2 = \mathbf{E}(y_1 + \cdots + y_n)^2 \rightarrow \infty$ as $n \rightarrow \infty$.

PROOF. It is obvious that

$$\sigma_n^2 = \mathbf{E}(y_1 + \cdots + y_n)^2 = \sum_{k=-\infty}^{\infty} |c_{k-1} + \cdots + c_{k-n}|^2.$$

We shall show that

$$(2.20) \quad \frac{|c_{kn}|}{\sigma_n} = \frac{|c_{k-1} + \cdots + c_{k-n}|}{\sigma_n} \leq a_n = \sqrt{\frac{4 \sum_{k=-\infty}^{\infty} |c_k|^2}{\sigma_n}} \left(1 + \frac{1}{2\sigma_n} \right)$$

⁵ Cf. [21].

under the hypotheses of the theorem. We have

(2.21)
$$\begin{aligned} \frac{c_{k-l,n}^2}{\sigma_n^2} &= \frac{c_{k-l-1}^2 + c_{k-n-l-1}^2}{\sigma_n^2} + \frac{2(c_{k-l-1} - c_{k-n-l-1})c_{k-n-l-1,n}}{\sigma_n^2} + \frac{c_{k-l-1,n}^2}{\sigma_n^2}, \\ \frac{c_{k-l+1,n}^2}{\sigma_n^2} &= \frac{c_{k-l}^2 + c_{k-n-l}^2}{\sigma_n^2} + \frac{2(c_{k-l} - c_{k-n-l})c_{k-n-l,n}}{\sigma_n^2} + \frac{c_{k-l,n}^2}{\sigma_n^2}, \\ &\dots\dots\dots, \\ \frac{c_{kn}^2}{\sigma_n^2} &= \frac{c_{k-1}^2 + c_{k-n-1}^2}{\sigma_n^2} + \frac{2(c_{k-1} - c_{k-n-1})c_{k-1,n}}{\sigma_n^2} + \frac{c_{k-l,n}^2}{\sigma_n^2}, \end{aligned}$$

and therefore, adding all the equations (2.21), we obtain

(2.22)
$$\frac{c_{kn}^2}{\sigma_n^2} \leq \frac{2 \sum_{-\infty}^{\infty} c_k^2}{\sigma_n^2} + \frac{4 \sum c_k^2}{\sigma_n} + \frac{c_{k-l-1,n}^2}{\sigma_n^2}.$$

Since by suitably choosing l , we can make

$$\frac{c_{k-l-1,n}^2}{\sigma_n^2}$$

arbitrarily small, (2.22) is equivalent to (2.20). Now, writing $c_{kn}/\sigma_n = a_{kn}$, we have

$$\frac{1}{\sigma_n} (y_1 + \dots + y_n) = \sum_{-\infty}^{\infty} a_{kn} x_k, \quad \sum_{-\infty}^{\infty} a_{kn}^2 = 1.$$

Let $\{\varepsilon_j\}$ be a sequence of numbers such that $\varepsilon_j \rightarrow 0$, and consider the sequence of series of independent random variables

$$\begin{aligned} &\xi_{11}, \xi_{12}, \dots, \xi_{1k_1}, \\ &\dots\dots\dots, \\ &\xi_{n1}, \xi_{n2}, \dots, \xi_{nk_n}, \\ &\dots\dots\dots, \end{aligned}$$

where

$$\xi_{n1} = \sum_{|k| > N} a_{kn} x_k,$$

$$\xi_{n2} = a_{-N,n} x_{-N}, \quad \xi_{n3} = a_{-N+1,n} x_{-N+1}, \dots, \quad \xi_{nk_n} = a_{Nn} x_N,$$

and N is chosen so that

$$\sum_{|k| > N} a_{kn}^2 < \varepsilon_n.$$

Then

$$\sum_{j=1}^{k_n} \int_{|\xi_{nj}| > \varepsilon} \xi_{nj}^2 \mathbf{P}(d\omega) \leq \int_{|x_0| > \varepsilon/a_n} x_0^2 \mathbf{P}(d\omega) + \varepsilon_n = o(1),$$

and hence (see [6], pp. 101–102)

$$\mathbf{P}\{\xi_{n1} + \dots + \xi_{nk_n} < z\} = \mathbf{P}\left\{\frac{y_1 + \dots + y_n}{\sigma\sqrt{n}} < z\right\} \rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

EXAMPLE. We remarked earlier that there exist stationary processes

generated by processes with strong mixing which are not themselves processes with strong mixing. We now give an example of such a process. Let $\{\varepsilon_j\}$ be a sequence of independent, identically distributed random variables, for which

$$\mathbf{P}\{\varepsilon_j = 1\} = \mathbf{P}\{\varepsilon_j = 0\} = 1/2.$$

Setting

$$\begin{aligned} & \dots\dots\dots, \\ x_1 &= \frac{\varepsilon_1}{2} + \frac{\varepsilon_2}{4} + \dots + \frac{\varepsilon_k}{2^k} + \dots, \\ & \dots\dots\dots, \\ x_j &= \frac{\varepsilon_j}{2} + \frac{\varepsilon_{j+1}}{4} + \dots + \frac{\varepsilon_{j+k}}{2^k} + \dots, \\ & \dots\dots\dots, \end{aligned}$$

we easily see that

$$\mathbf{E}|x_1 - \mathbf{E}\{x_1 | \varepsilon_1, \dots, \varepsilon_k\}|^2 \leq 2^{-k}.$$

If the process $\{x_j\}$ had the strong mixing property, then any process of the form $\{f(x_j)\}$ would have the same property. In this case,

$$\mathbf{P}\left\{\frac{\sum_1^n f(x_j)}{B_n} < z\right\} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du$$

would imply (Theorem 1.1) that $B_n^2 = nh(n)$, where $h(n)$ is a slowly varying function. Let

$$f_1 = f(x_1) = \sum_{k=1}^{\infty} r_k(x_1) k^{-3/4},$$

where $r_k(x_1)$ is the Rademacher function:

$$r_k(x_1) = \text{sign} \sin 2^k \pi x_1, \quad r_k = -1 + 2\varepsilon_k.$$

The random variables r_k are independent and have zero mean, and hence

$$\begin{aligned} \mathbf{E}f(x_1)f(x_{j+1}) &= \mathbf{E}f_1f_{j+1} = \sum_{k=1}^{\infty} k^{-3/4}(k+j)^{-3/4} > \frac{2^{-3/4}}{j^{3/4}}, \\ \sigma_n^2 &= \mathbf{E}\left(\sum_1^n f_j\right)^2 = n\mathbf{E}f_1^2 + 2\sum_{k=1}^n (n-k)\mathbf{E}f_0f_k > n^{5/4}(1+o(1)). \end{aligned}$$

The stationary process $\{f(x_j)\}$ satisfies all the hypotheses of the preceding theorem, and hence

$$\mathbf{P}\left\{\frac{\sum_1^n f_j}{\sigma_n} < z\right\} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

Obviously, $\sigma_n^2 > n^{5/4}(1+o(1))$ cannot be represented in the form $nh(n)$, where $h(n)$ is a slowly varying function, and therefore the process $\{f_j\}$ does not have the strong mixing property.

2.2. A Refinement of McMillan's Theorem

Let $\{x_j\}$ be a stationary random process with a finite number of states, i.e., suppose the random variable x_0 takes only a finite set of values $a_1, a_2, \dots, a_m$ with probability 1. We define a random variable $p(x_1, \dots, x_n)$ in the following way:

$$p(x_1, \dots, x_n) = \mathbf{P}\{x_1 = a_{i_1}, \dots, x_n = a_{i_n}\} \quad \text{if } x_1 = a_{i_1}, \dots, x_n = a_{i_n}.$$

If the process $\{x_j\}$ is ergodic, then the quantity ⁶

$$-\frac{\log p(x_1, \dots, x_n)}{n}$$

converges in probability to the entropy H of the process $\{x_j\}$. This theorem due to McMillan (see [20], [28]) implies the information stability of an important class of messages [9], and is needed to prove certain theorems of information theory (see e.g., [28]). In [29] A. A. Yushkevich has shown that if the quantities $\{x_j\}$ are connected in a simple finite, stationary ergodic Markov chain, then the distribution function of the quantity

$$\frac{1}{\sqrt{n}} (\log p(x_1, \dots, x_n) + nH)$$

is asymptotically normal. Here we shall prove a theorem which generalizes this result of A. A. Yushkevich (cf. also [24]).

Theorem 2.6. *Let $\{x_j\}$ be a stationary process with a finite number of states $a_1, a_2, \dots, a_m$ which satisfies the strong mixing condition (II), and suppose that*

1. $\sum_{n=1}^{\infty} |\alpha(n)|^{1-\delta_1} < \infty \quad \text{for some } \delta_1 > 0,$
2. $\sum_{n=1}^{\infty} |\psi(n)|^{(1-\delta_2)/2} < \infty \quad \text{for some } \delta_2 > 0,$

where

$$\psi(n) = \sup_{1 \leq i \leq m} |\mathbf{E}[\mathbf{P}(x_0 = a_i | x_1, \dots) - \mathbf{P}(x_0 = a_i | x_1, \dots, x_n)]|.$$

Then

$$\mathbf{P} \left\{ \frac{\log p(x_1, \dots, x_n) + nH}{\sqrt{n}} < z \right\} \xrightarrow{n \rightarrow \infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z/\sigma} e^{-u^2/2} du,$$

where

$$H = -\mathbf{E}(\log p(x_0 | x_{-1}, \dots))$$

is the entropy of the process $\{x_j\}$,

$$\begin{aligned} \sigma^2 &= \mathbf{E}(\log p(x_0 | x_{-1}, \dots) + H)^2 \\ &\quad + 2 \sum_{i=1}^{\infty} \mathbf{E}(\log p(x_0 | x_{-1}, \dots) + H)(\log p(x_i | x_{i-1}, \dots) + H), \end{aligned}$$

⁶ Logarithms are taken to the base 2 everywhere in Section 2.2.

and the random variable $p(x_i|x_{i-1}, x_{i-2}, \dots)$ is defined by the relation

$$p(x_i|x_{i-1}, x_{i-2}, \dots) = \mathbf{P}\{x_i = a_j|x_{i-1}, \dots\} \quad \text{if } x_i = a_j.$$

PROOF. If

$$f_s = \log p(x_s|x_{s-1}, \dots), \quad g_s = \log p(x_s|x_{s-1}, \dots, x_1),$$

then

$$\mathbf{E} \left| \frac{\log p(x_1, \dots, x_n)}{\sqrt{n}} - \frac{\sum_1^n f_s}{\sqrt{n}} \right| \leq \frac{1}{\sqrt{n}} \sum_1^n \mathbf{E}|f_s - g_s| = o(1)$$

because of condition 2 of the theorem. Thus, it is sufficient to exhibit the asymptotic normality of the sums

$$\frac{1}{\sqrt{n}} \sum_{s=1}^n f_s.$$

To achieve this, we use Theorem 2.2. Condition 3 of this theorem is met because of condition 1 of Theorem 2.6. We must still verify that conditions 1 and 2 are met.

Lemma 2.1. *For any $r > 0$ whatsoever, we have*

$$\mathbf{E}|f_0|^r < \infty.$$

PROOF. Let A_{il} be the event consisting of

$$l \leq |\log \mathbf{P}\{x_0 = a_i|x_{i-1}, \dots\}| < l+1, \quad l = 1, 2, \dots$$

The event A_{il} is equivalent to the event

$$2^{-l} \leq \mathbf{P}\{x_0 = a_i|x_{i-1}, \dots\} < 2^{-(l+1)}.$$

Therefore

$$P\{(x_0 = a_i) \cap A_{il}\} \leq 2^{-l}$$

and hence

$$\mathbf{E}|f_0|^r \leq m \sum_{l=1}^{\infty} (l+1)^r 2^{-l} < \infty.$$

Lemma 2.2. *For any $r \geq 2$ and any $\delta > 0$,*

$$\mathbf{E}|f_0 - \mathbf{E}\{f_0|x_0, \dots, x_{-k}\}|^r \leq K(r, \delta) |\psi(k)|^{(1-\delta)/2},$$

where $K(r, \delta)$ is a constant.

PROOF. First let $r = 2$. It is obvious that

$$\begin{aligned} \mathbf{E}|f_0 - \mathbf{E}\{f_0|x_0, \dots, x_{-k}\}|^2 &\leq \mathbf{E}|f_0 - \log p(x_0|x_{-1}, \dots, x_{-k})|^2 \\ (2.23) \quad &= \sum_{i=1}^m \int_{\{x_0=a_i\}} \log^2 \frac{\mathbf{P}\{x_0 = a_i|\mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{x_0 = a_i|\mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega). \end{aligned}$$

Let A_i denote the event $\{x_0 = a_i\}$. The relation (2.23) allows us to consider only the estimation of the integrals

$$\int_{A_i} \left| \log \frac{\mathbf{P}\{x_0 = a_i | \mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{x_0 = a_i | \mathfrak{M}_{-\infty}^{-1}\}} \right|^2 \mathbf{P}(d\omega).$$

To obtain the necessary estimates, we decompose Ω into non-intersecting events. If

$$\theta_{ik} = \mathbf{P}\{A_i | \mathfrak{M}_{-k}^{-1}\} - \mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\},$$

then

$$\int_{A_i} \log^2 \frac{\mathbf{P}\{A_i | \mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega) = \int_{A_i} \log^2 \left(1 + \frac{\theta_{ik}}{\mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}} \right) \mathbf{P}(d\omega).$$

Let

$$A_{il}^+ = \left\{ \omega : 2^l \leq 1 + \frac{\theta_{ik}}{\mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}} < 2^{l+1} \right\}, \quad l = 1, 2, \dots$$

We have

$$(2.24) \quad 2^l \mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\} \leq \mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\} + \theta_{ik} \leq 2^{l+1} \mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}.$$

Obviously, $A_{il}^+ \in \mathfrak{M}_{-\infty}^{-1}$; integrating (2.24) over the set A_{il}^+ , we find that

$$(2.25) \quad \mathbf{P}\{A_i A_{il}^+\} \leq \frac{\mathbf{E}|\theta_{ik}|}{2^l - 1}.$$

Moreover, let

$$A_{il}^- = \left\{ \omega : 2^{-l} \geq 1 + \frac{\theta_{ik}}{\mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}} > 2^{-l-1} \right\}.$$

From the inequality

$$2^{-l} \mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\} \geq \mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\} + \theta_{ik} > 2^{-l-1} \mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}$$

which is satisfied on A_{il}^- , we find, arguing as before, that

$$(2.26) \quad \mathbf{P}\{A_i A_{il}^-\} \leq \frac{2^l}{2^l - 1} \mathbf{E}|\theta_{ik}|.$$

At the same time,

$$\begin{aligned} A_{il}^- &= \left\{ \omega : -l \geq \log \frac{\mathbf{P}\{A_i | \mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}} > -l-1 \right\} \\ &\subset \left\{ \omega : -l \geq \log \frac{\mathbf{P}\{A_i | \mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i | \mathfrak{M}_{-\infty}^{-1}\}} \right\} \cup \left\{ \omega : -l \geq \log \mathbf{P}\{A_i | \mathfrak{M}_{-k}^{-1}\} \right\} = B_{il}. \end{aligned}$$

Obviously $\mathbf{P}\{A_i | \mathfrak{M}_{-k}^{-1}\} \leq 2^{-l}$ on B_{il} , and moreover,

$$B_{il} \in \mathfrak{M}_{-k}^{-1} \text{ and } \mathbf{P}\{A_{il}^- A_i\} \leq \mathbf{P}\{B_{il} A_i\},$$

so that

$$(2.27) \quad \mathbf{P}\{A_{il}^- A_i\} \leq \mathbf{P}\{B_{il} A_i\} \leq 2^{-l}.$$

We determine the integer $l = l(\delta)$ from the condition $l\delta < 1 - \delta < (l+1)\delta$.

Now suppose that for $\psi(k) < 1$

$$\begin{aligned}
 {}_jA_{i\delta}^+ &= \left\{ \omega : |\psi(k)|^{(j+1)\delta} \leq \log \left(1 + \frac{\theta_{ik}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \right) < |\psi(k)|^{j\delta} \right\}, \quad j = 0, 1, \dots, l. \\
 (2.28) \quad {}_jA_{i\delta}^- &= \left\{ \omega : -|\psi(k)|^{j\delta} \leq \log \left(1 + \frac{\theta_{ik}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \right) < -|\psi(k)|^{(j+1)\delta} \right\},
 \end{aligned}$$

$$(2.29) \quad A_{i\delta} = \left\{ \omega : -|\psi(k)|^{(l+1)\delta} \leq \log \left(1 + \frac{\theta_{ik}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \right) \leq |\psi(k)|^{(l+1)\delta} \right\}.$$

Since

$$2^{|\psi(k)|(j+1)\delta} \leq 1 + \frac{\theta_{ik}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \leq 2^{|\psi(k)|j\delta}$$

on ${}_jA_{i\delta}^+$, then

$$(2.30) \quad \mathbf{P}\{A_i \cap {}_jA_{i\delta}^+\} \leq \frac{\mathbf{E}|\theta_{ik}|}{2^{|\psi(k)|(j+1)\delta} - 1} = M|\psi(k)|^{1-(j+1)\delta}.$$

Similarly,

$$\mathbf{P}\{A_i \cap {}_jA_{i\delta}^-\} = M|\psi(k)|^{1-(j+1)\delta}.$$

Moreover, we write

$$\begin{aligned}
 \int_{A_i} \log^2 \frac{\mathbf{P}\{A_i|\mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega) &= \left[\int_{\cup_{l=1}^{\infty} A_i A_{il}^+} + \int_{\cup_{l=1}^{\infty} A_i A_{il}^-} \right. \\
 &\quad \left. + \sum_{j=0}^l \left(\int_{{}_jA_{i\delta}^+} + \int_{{}_jA_{i\delta}^-} \right) + \int_{A_{i\delta}} \right] \log^2 \frac{\mathbf{P}\{A_i|\mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega).
 \end{aligned}$$

The inequality (2.25) gives

$$\int_{\cup_{l=1}^{\infty} A_i A_{il}^+} \log^2 \frac{\mathbf{P}\{A_i|\mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega) \leq \sum_{l=1}^{\infty} \frac{(l+1)^2 \psi(k)}{2^l - 1} = M\psi(k).$$

From (2.26) and (2.27), we obtain

$$\begin{aligned}
 \int_{\cup_{l=1}^{\infty} A_i A_{il}^-} \log^2 \frac{\mathbf{P}\{A_i|\mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega) &\leq \sum_{l=1}^{[\log \psi(k)]} \frac{(l+1)^2 2^l}{2^l - 1} \psi(k) \\
 &\quad + \sum_{[\log \psi(k)]+1}^{\infty} (l+1)^2 2^{-l} = M|\log \psi(k)|^3 \psi(k).
 \end{aligned}$$

Moreover, from (2.28)–(2.29) and (2.30), we have

$$\begin{aligned}
 \int_{{}_jA_{i\delta}^+} \log^2 \frac{\mathbf{P}\{A_i|\mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega) &= M|\psi(k)|^{1-\delta}, \\
 \int_{A_{i\delta}} \log^2 \frac{\mathbf{P}\{A_i|\mathfrak{M}_{-k}^{-1}\}}{\mathbf{P}\{A_i|\mathfrak{M}_{-\infty}^{-1}\}} \mathbf{P}(d\omega) &\leq |\psi(k)|^{(l+1)\delta} = M(\psi(k))^{1-\delta},
 \end{aligned}$$

and hence

$$\mathbf{E}|f_0 - \mathbf{E}\{f_0|x_0, \dots, x_{-k}\}|^2 \leq K|\psi(k)|^{1-\delta}.$$

In the general case, if $r > 2$

$$\begin{aligned} \mathbf{E}|f_0 - \mathbf{E}\{f_0|x_0, \dots, x_{-k}\}|^r &\leq \mathbf{E}^{1/2}|f_0 - \mathbf{E}\{f_0|x_0, \dots, x_{-k}\}|^2 \\ &\times \mathbf{E}^{1/2}|f_0 - \mathbf{E}\{f_0|x_0, \dots, x_{-k}\}|^{2(r-1)} \leq K|\psi(k)|^{(1-\delta)/2}, \end{aligned}$$

where, of course, the constant K depends on r and δ . This proves the lemma.

We now choose r and δ such that they satisfy the inequalities

$$\frac{r-2}{r} > 1-\delta_1, \quad \frac{r-1}{r}(1-\delta) > 1-\delta_2.$$

Then

$$\sum_{n=1}^{\infty} |\alpha(n)|^{(r-2)/r} = M \sum_{n=1}^{\infty} |\alpha(n)|^{1-\delta_1} < \infty,$$

and

$$\begin{aligned} &\sum_{n=1}^{\infty} \mathbf{E}^{(r-1)/r} \{ |f_0 - \mathbf{E}\{f_0|x_{-1}, \dots, x_{-n}\}|^{r/(r-1)} \} \\ &= M \sum_{n=1}^{\infty} |\psi(n)|^{((r-1)/r)(1-\delta)/2} = M \sum_{n=1}^{\infty} |\psi(n)|^{(1-\delta_2)/2} < \infty, \end{aligned}$$

because of Lemma 2.2. Thus, for the process $\{f_j\}$ constructed from the random variable $f = \log p(x_0|x_{-1}, \dots)$ in the way indicated in Theorem 2.2, all the conditions of this theorem are satisfied. Moreover,

$$\begin{aligned} \mathbf{E}f_0 &= \mathbf{E} \log p(x_0|x_{-1}, \dots) = \lim_{n \rightarrow \infty} \mathbf{E} \log p(x_0|x_{-1}, \dots, x_{-n}) \\ &= \lim_{n \rightarrow \infty} \mathbf{E}(\log p(x_{n+1}, \dots, x_0) - \log p(x_n, \dots, x_0)) \\ &= \lim_{n \rightarrow \infty} \frac{\mathbf{E} \log p(x_0, \dots, x_n)}{n} = -H, \end{aligned}$$

which means that the sums

$$\frac{1}{\sigma\sqrt{n}} \left(\sum_{j=1}^n f_j + nH \right),$$

and with them the quantities

$$\frac{1}{\sigma\sqrt{n}} (\log p(x_1, \dots, x_n) + nH),$$

are asymptotically normal with mean 0 and variance 1. This proves the theorem.

REMARK. Theorem 2.6 can be strengthened somewhat by replacing condition 2 by the condition

$$2'. \quad \sum_n |\psi(n)|^{1-\delta_2} < \infty \text{ for some } \delta_2 > 0.$$

This strengthening is based on the fact that we prove the asymptotic normality of the sums

$$S_n = \frac{1}{\sigma\sqrt{n}} \sum_{j=1}^n f_j + nH$$

not by using Theorem 2.2, but directly, by setting

$$f_j = \log p(x_j | x_{j-1}, \dots) = \xi_j^{(s)} + \eta_j^{(s)},$$

where

$$\xi_j^{(s)} = p(x_j | x_{j-1}, \dots, x_{j-s}), \quad \eta_j^{(s)} = f_j - \xi_j^{(s)},$$

and then argue as in the proof of Theorem 2.2.

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SOME LIMIT THEOREMS FOR STATIONARY PROCESSES

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(Summary)

In this paper stationary stochastic processes in the strong sense are investigated, which satisfy the condition

$$|\mathbf{P}(AB) - \mathbf{P}(A)\mathbf{P}(B)| \leq \varphi(n)\mathbf{P}(A), \quad \varphi(n) \downarrow 0,$$

for every $A \in \mathfrak{M}_{-\infty}^0$, $B \in \mathfrak{M}_n^\infty$, or the “strong mixing condition”

$$\sup_{A \in \mathfrak{M}_{-\infty}^0, B \in \mathfrak{M}_n^\infty} |\mathbf{P}(AB) - \mathbf{P}(A)\mathbf{P}(B)| \alpha(n) \downarrow 0,$$

where $\mathfrak{M}_a^b$ is a σ -algebra generated by the events

$$\{(x_{i_1}, x_{i_2}, \dots, x_{i_k}) \in \mathbf{E}\}, \quad a \leq i_1 < i_2 < \dots < i_k \leq b,$$

$\mathbf{E}$ being a k -dimensional Borel set.

Some limit theorems for the sums of the type

$$\frac{x_1 + \dots + x_n}{B_n} - A_n \quad \text{or} \quad \frac{f_1 + \dots + f_n}{B_n} - A_n$$

are established. Here $f_j = T^j f$, and the random variable f is measurable with respect to $\mathfrak{M}_{-\infty}^\infty$.